

Activity: Library Book Management System

Learning Outcomes

By completing this activity, students will be able to:

- Create and use **constructors**.
 - Store objects in an **array**.
 - Use **pointers** to access and manipulate object data.
 - Apply searching techniques to locate records.
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Problem Statement

A library wants a simple system to manage its books.

Create a C++ program using a class named **Book**.

The class should contain the following private data members:

- Book ID
- Book Title
- Author Name
- Number of Copies

Requirements

Part A – Constructor (20 Marks)

Create:

- A **default constructor** that initializes all data members to default values.
 - A **parameterized constructor** that accepts all book information.
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Part B – Array of Objects (20 Marks)

Declare an array that can store information for **5 books**.

Initialize the array using the parameterized constructor.

Part C – Pointer Operations (30 Marks)

Create a pointer that points to the first object in the array.

Using **pointer arithmetic only** (`ptr + i` and `->` operator), display:

- Book ID
- Book Title
- Author Name
- Number of Copies

Do not access the objects using array indexing (`books[i]`) in this section.

Part D – Search Book (15 Marks)

Prompt the user to enter a **Book ID**.

Using the pointer, search for the book.

If found, display all book information.

Otherwise, display:

```
Book not found.
```

Part E – Update Copies (15 Marks)

After a successful search:

- Ask the user to enter the number of newly arrived copies.
 - Update the number of copies using the pointer.
 - Display the updated book information.
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Sample Output

```
Library Books
```

```
Book ID: B101
Title: Data Structures
Author: Mark Allen
Copies: 8
```

```
Book ID: B102
Title: C++ Programming
Author: John Smith
Copies: 12
```

```
Book ID: B103
Title: Database Systems
Author: Peter Brown
Copies: 5
```

Book ID: B104
Title: Operating Systems
Author: Linda Tan
Copies: 9

Book ID: B105
Title: Computer Networks
Author: David Lee
Copies: 7

Enter Book ID to search: B103

Book Found
Book ID: B103
Title: Database Systems
Author: Peter Brown
Copies: 5

Enter additional copies: 3

Updated Book Information
Book ID: B103
Title: Database Systems
Author: Peter Brown
Copies: 8